

Can large language models meet the challenge of generating school-level questions?

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ABSTRACT

In the realm of education, crafting appropriate questions for examinations is a meticulous and time-consuming task that is crucial for assessing students' understanding of the subject matter. This paper explores the potential of leveraging large language models (LLMs) to automate question generation in the educational domain. Specifically, we focus on generating educational questions from contexts extracted from school-level textbooks. Our study aims to prompt LLMs such as GPT-4 Turbo, GPT-3.5 Turbo, Llama-2-70B, Llama-3.1-405B, and Gemini Pro to generate a complete set of questions for each context, potentially streamlining the question generation process for educators. We performed a human evaluation of the generated questions, assessing their *coverage*, *grammaticality*, *usefulness*, *answerability*, and *relevance*. Additionally, we prompted LLMs to generate questions based on Bloom's revised taxonomy, categorizing and evaluating these questions according to their cognitive complexity and learning objectives. We applied both zero-shot and eight-shot prompting techniques. These efforts provide insight into the efficacy of LLMs in automated question generation and their potential in assessing students' cognitive abilities across various school-level subjects. The results show that employing an eight-shot technique improves the performance of human evaluation metrics for the generated complete set of questions and helps generate questions that are better aligned with Bloom's revised taxonomy.

1. Introduction

In the realm of education, crafting appropriate questions for examinations is a meticulous and time-consuming task for educators (Gong et al., 2022; Kurdi et al., 2020; Wang et al., 2024). Developing effective questions requires careful consideration of various factors, including students' cognitive abilities, learning objectives, and curriculum standards (Al Faraby et al., 2023). The quality and diversity of the questions significantly impact the effectiveness of assessments in evaluating the students' understanding of the subject matter (Rahman et al., 2011; Howard & Aleman, 2008; Wilson & Floden, 2003). Educators must ensure that the questions align with the knowledge levels of their students while addressing several key issues such as *coverage* (Nenkova & Passonneau, 2004), *grammatical* precision (Maity et al., 2024a, 2024b), *usefulness* (Elkins et al., 2023; Mulla & Gharpure, 2023), *answerability* (Steuer et al., 2021; Wang et al., 2024), and *relevance* (Steuer et al., 2021; Mulla & Gharpure, 2023; Wang et al., 2024) to the subject matter. In addition, educators must cultivate higher-order thinking skills, such as analysis, application, and evaluation, rather than simply encouraging the recall of facts (Billing, 2007; Jensen et al., 2014). By emphasizing these di-

mensions, we gain a clearer understanding of the complexities involved in generating effective school-level questions, thus enriching the educational experience. This process requires a deep understanding of the strengths, weaknesses, and prior knowledge of the students (Birenbaum & Dochy, 2012). Furthermore, questions must align with the curriculum standards and learning objectives set by educational authorities (Drake & Burns, 2004). Ensuring that questions cover the required material without being too broad or too narrow can be a delicate balance. Good questions should go beyond simple recall of facts and encourage critical thinking, analysis, and problem-solving skills. Crafting questions that stimulate these higher-order thinking skills requires creativity and careful planning (Fisher & Frey, 2014).

As technology continues to advance, there is a potential solution to alleviate the burden on educators: leveraging large language models (LLMs) for automated generation of educational questions (Kasneeci et al., 2023). These models have the potential to alleviate some of the burdens educators face by providing a systematic approach to generating questions based on the content of school-level textbooks (Maity et al., 2024a). In this paper, we delve into the realm of automated question generation (AQG) using LLMs, specifically focusing on their applica-

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tion in the educational domain. Our study revolves around experimenting with generating school-level questions using context extracted from school-level textbooks spanning various subjects. Our primary objective is to leverage LLMs to generate a ‘complete set of questions’ for each context extracted from school-level textbooks. We define a ‘complete set of questions’ as a collection that thoroughly addresses all critical aspects of the given context, ensuring comprehensive coverage of the context without leaving significant gaps. This set includes questions ranging from basic factual recall to more complex conceptual questions, providing a well-rounded assessment of the subject matter corresponding to the context. Completeness is determined not by the number of questions but by their capacity to cover the essential details of the topic in a balanced and comprehensive manner. By harnessing the capabilities of advanced LLMs, we aim to streamline this process, reducing the time and effort required by educators. These models enable the generation of questions that span a wide range of content dimensions, from surface-level recall to deeper conceptual understanding, making the task of creating comprehensive question sets more efficient. Unlike Bloom’s revised taxonomy (Bloom, 2010), which classifies questions according to cognitive complexity (Stayanchi, 2018), we define a ‘complete set of questions’ based on how comprehensively the questions address the content. While Bloom’s revised taxonomy is valuable for categorizing cognitive levels, our focus on completeness emphasizes ensuring that all key dimensions of the context are covered (Al Faraby et al., 2023). To assess the quality of the generated questions, we conduct a human evaluation based on several key metrics: *coverage*, *grammar*, *usefulness*, *answerability*, and *relevance*. This evaluation ensures that the generated questions are not only comprehensive but also meet the standards necessary for effective educational assessment.

Additionally, we prompted LLMs to generate questions based on Bloom’s revised taxonomy, categorizing and evaluating the generated questions according to their cognitive complexity and learning objectives. Through this dual analysis, we aim to illuminate the variety and depth of questions produced by LLMs. By examining both the cognitive diversity and content completeness of the generated questions, we provide insights into the efficacy of LLMs in automating the generation of educational questions. The contributions of our work are as follows:

- We used zero-shot prompting with LLMs to generate a *complete set of questions* from contexts extracted from school-level textbooks on several subjects (See Section 4.1). Subsequently, we performed a human evaluation of the generated questions based on several human evaluation metrics (See Section 4.3). Additionally, we generated questions by providing the LLMs with eight-shot examples.
- We also prompted LLMs to generate questions based on Bloom’s revised taxonomy (See Section 4.2), and then analyzed the generated questions through human experts in both zero-shot and eight-shot settings.

2. Related work

Automated question generation (AQG) has seen a rapid evolution, transitioning from early rule-based approaches to sophisticated methods leveraging LLMs. This section provides a holistic review of the key developments in the field, with an emphasis on their pedagogical applications and limitations. The discussion begins with foundational rule-based methods, highlighting their constrained adaptability. It then progresses to machine learning techniques, showcasing their contribution to contextual relevance and domain specificity, before examining the transformative role of LLMs in AQG. Finally, advancements in prompt engineering and the integration of pedagogical frameworks like Bloom’s revised taxonomy are explored to identify persistent challenges and opportunities for improvement. This review concludes by detailing how our work addresses the educational needs in AQG, offering solutions to enhance learning outcomes and streamline question generation processes.

The earliest AQG systems predominantly relied on rule-based methods (Kurdi et al., 2020; Mulla & Gharpure, 2023), which utilized pre-defined templates and syntactic rules to generate questions. These systems, exemplified by the work of Montenegro et al. (2012), were particularly effective in structured environments but exhibited significant limitations in scalability and adaptability. For instance, these methods required manual crafting of domain-specific rules, making them labor-intensive and unable to address diverse educational needs. While effective in generating grammatically correct questions, rule-based systems often failed to capture the contextual depth and semantic nuances necessary for meaningful learning assessments.

The introduction of machine learning (ML) models marked a paradigm shift in AQG, enabling the generation of more contextually relevant and domain-specific questions. Various ML techniques, including multi-task learning (Wang et al., 2017), reinforcement learning (Fan et al., 2018), graph neural networks (GNN) (Chen et al., 2019), and variational autoencoders (Wang et al., 2019), demonstrated significant advancements. Pre-trained models (Dong et al., 2019) and transfer learning (Liao & Koh, 2020) further enhanced the field by enabling knowledge transfer across domains. Despite these innovations, ML-driven AQG systems faced notable challenges. Many struggled to generate high-quality questions due to biases in training data, which often skewed outputs toward specific patterns or domains (Parsia et al., 2015). Additionally, the specialized knowledge required for subject-specific questions was difficult to encode, limiting their applicability in diverse educational settings (Liao & Koh, 2020). These shortcomings underscored the necessity for AQG approaches that are both flexible and tailored to educators’ pedagogical needs.

In recent years, transformer-based LLMs have emerged as a dominant force in AQG, particularly within the realm of educational technology (Hao et al., 2024). These models, entrenched in machine learning, are trained on extensive and diverse datasets, significantly enhancing their generative capabilities (Zhang et al., 2023). The rationale behind employing LLMs in AQG lies in their superior ability to produce contextually relevant and diverse questions, compared to traditional rule-based systems and earlier AI-driven methodologies (Kurdi et al., 2020; Mulla & Gharpure, 2023; Steuer et al., 2021). For instance, rule-based systems, while effective in structured environments, often fail to capture the nuance and complexity needed for educational assessment. In contrast, LLMs are capable of generating a wide range of question types, thus better meeting the varied needs of education (Al Faraby et al., 2024). The primary goal in training transformer-based LLMs is often to predict the next token, enabling these models to generate text that naturally follows a given input. With recent developments, training methods now increasingly incorporate reinforcement learning, adding an additional dimension to LLM training. This advancement is particularly evident in models such as GPT-4 Turbo (OpenAI, 2023), GPT-3.5 Turbo, Llama-2-70B (Touvron et al., 2023), Llama-3.1-405B (Dubey et al., 2024), and Gemini Pro (Team et al., 2023), all of which are used in the experiments described in this article. Through reinforcement learning and feedback from human evaluators, these models achieve greater performance than earlier LLMs (OpenAI, 2023; Touvron et al., 2023; Ouyang et al., 2022).

Following the standard LLM training aim of predicting subsequent tokens, recent AQG strategies use textual prompts provided to LLMs to elicit generated responses or completions (Mulla & Gharpure, 2023). The design of these prompts, aimed at shaping the generated output, has spurred a growing field of research known as prompt engineering. A common approach in prompt engineering is the prefix-style prompt, where a specific string is added at the beginning of the text given to the LLM to set the generation context (Liu et al., 2023). For instance, a biology teacher looking to create questions about photosynthesis might start with a basic prompt: “Generate a question about photosynthesis”. To achieve more detailed or precise questions, the teacher could refine the prompt by adding context, such as a textbook excerpt that discusses particular aspects of photosynthesis, using a more targeted prompt like: “Given the context <context>, generate a Question” (Maity et al., 2024a).

Context: Purchasing power parity (PPP) is an economic indicator that signifies the purchasing power of the currencies of various nations of the world against each other. It helps in comparing living standards between different countries and estimating economic productivity.
Long Prompt: purchasing power parity helps
Short Prompt: purchasing power
Gold Standard Question : What does purchasing power parity do?

Fig. 1. A sample from the EduProbe (Maity et al., 2024a) dataset, illustrating content aligned with the 11th standard Economics syllabus under the NCERT curriculum.

For even greater control over the question generation, the teacher could integrate guiding elements, such as specific keywords that direct the type of question, by using prompts like: “Generate multiple-choice questions based on the given context <context> along with the correct answer and three distractors” (Maity et al., 2024b).

A growing body of literature highlights the efficacy of AI-driven AQG systems in educational contexts. For example, Vachev et al. (2022) introduced the Leaf AQG system, which fine-tunes T5 (Raffel et al., 2020) for question generation and multiple-choice distractor creation. Leaf makes use of the SQuAD dataset (Rajpurkar et al., 2016), which contains more than 100,000 questions drawn from diverse Wikipedia articles, to train the AQG system. Muse et al. (2023) proposed the educational multiple-choice question generation model EduQG by fine-tuning the T5 model on the S2ORC dataset (Lo et al., 2020), which contains 23.2M scientific abstracts related to biology, chemistry, and physics research papers, followed by automatic and human evaluation. Fawzi et al. (2023) prompted ChatGPT to generate educational multiple-choice questions. The prompt they used was: “Given text <context>, create 5 expert level questions with multiple choice answers from the text”. They used the S2ORC dataset for their experiments. However, none of the studies explored Bloom’s revised taxonomy for analyzing the generated multiple-choice questions.

Another aspect of prompt engineering is including examples within the prompt that illustrate the desired format and style. This approach, called few-shot learning, typically combines an instruction with a few sample instances that illustrate the task. These examples allow LLMs to adapt to new contexts without additional training or fine-tuning (Liu et al., 2023). In their study, Wang et al. (2022) explored various prompting techniques for generating educational questions, finding that shorter input contexts along with few-shot learning resulted in higher-quality question outputs. Their research focused specifically on the GPT-2 model, using the SQuAD dataset and OpenStax textbooks as sources (Stafford & Flatley, 2018). Additionally, Bhandari et al. (2023) studied question generation using the OpenStax College Algebra textbook (Abramson, 2021) as a resource, employing ChatGPT in a 0-shot setting to streamline the question generation process. The 0-shot prompt they utilized was: “Please generate 20 exercise questions based on this textbook chapter.” Lee et al. (2024) also explored zero-shot and few-shot learning for automated question generation in English education. However, no studies have yet incorporated Bloom’s revised taxonomy to classify and evaluate the generated questions.

Despite these advancements, a critical gap exists in systematically assessing the quality of AI-generated questions using established pedagogical frameworks, particularly Bloom’s revised taxonomy. This framework classifies cognitive processes and aligns them with educational objectives, providing a robust foundation for evaluating the appropriateness of questions across various levels of learning (Bloom, 2010; Stayanchi, 2018). Many existing studies (Maity et al., 2024a; Muse et al., 2023; Fawzi et al., 2023; Maity et al., 2024b) have not integrated this framework into their AQG systems, nor have they developed comprehensive methods for assessing the completeness and relevance of the generated questions. In our work, we define a ‘complete set of questions’ as a collection that thoroughly covers all critical aspects of a context extracted from school-level textbooks, emphasizing comprehensive detail over mere quantity. By leveraging advanced LLMs, we aim to generate

questions that span from basic factual recall to more complex conceptual inquiries, ultimately enhancing learning outcomes. This study investigates the ability of state-of-the-art LLMs to generate a ‘complete set of questions’ from an educational context that meets pedagogical standards and aligns with Bloom’s revised taxonomy. The implications for practice were profound. Educators equipped with AI-driven AQG tools could more efficiently design assessments that addressed diverse learning objectives, ultimately enhancing the learning experience by offering personalized and contextually relevant questions.

Building upon prior advancements, our work introduced several key distinctions. Firstly, rather than relying on crowd-sourced data from sources like Wikipedia (e.g., SQuAD), we explored AQG from school-level educational textbooks covering subjects such as History, Geography, Economics, Environmental Studies, and Science. Secondly, we evaluated state-of-the-art LLMs, including GPT-4 Turbo, GPT-3.5 Turbo, Llama-2-70B, Llama-3.1-405B, and Gemini Pro, in both zero-shot (0-shot) and eight-shot (8-shot) settings, providing valuable insights into their performance across different configurations. Lastly, we investigated whether these LLMs were capable of generating a ‘complete set of questions’ from a context, along with AQG according to Bloom’s revised taxonomy, a dimension not explored in prior research.

3. Dataset

In this study, we used the EduProbe dataset introduced by Maity et al. (2024a), sourced from the National Council of Educational Research and Training (NCERT)¹ textbooks. This dataset covers various subjects, including history, geography, economics, environmental studies, and science, with content designed for students from 6th to 12th grade. Each entry in the dataset consists of four elements: 1) *Context*, the passage used as the basis for generating questions; 2) *Long Prompt*, an extended textual cue related to the main topic of the context; 3) *Short Prompt*, a concise textual cue that provides a brief focus, often derived from an initial noun phrase within the context; and 4) *Question*, a well-aligned question based on the context and prompts. The EduProbe dataset includes 3,502 <Context, Long Prompt, Short Prompt, Question> quadruples in total. Fig. 1 illustrates an example. From this dataset, we extracted 3,502 <Context, Question> pairs for our experiments.

4. Methodology

In this section, we outline the methodology used to evaluate the effectiveness of two proposed educational question generation methods (Sections 4.1 and 4.2) utilizing state-of-the-art LLMs, including the evaluation criteria (Section 4.3) and recruitment details for the teacher evaluators (Section 4.4). The two methods under scrutiny are designed to generate a complete set of questions from a provided context (Method 1) and to generate questions aligned with Bloom’s revised taxonomy (Method 2). Our investigation involves five prominent LLMs: GPT-4 Turbo, GPT-3.5 Turbo, Llama-2-70B, Llama-3.1-405B, and Gemini Pro.

¹ https://en.wikipedia.org/wiki/National_Council_of_Educational_Research_and_Training.

```

Generate a complete set of questions.

Context: {context}
Question 1:
Question 2:
Question 3:
Question 4:
[...]

```

Fig. 2. Zero-shot prompt template for generating a complete set of questions.

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Generate a complete set of questions. Please take care of all the examples given below
while generating the questions.

Context: {example_context}
Question 1: {example_question}
Question 2: {example_question}
Question 3: {example_question}
Question 4: {example_question}
[...]

Context: {example_context}
Question 1: {example_question}
Question 2: {example_question}
Question 3: {example_question}
Question 4: {example_question}
[...]

Context: {example_context}
Question 1: {example_question}
Question 2: {example_question}
Question 3: {example_question}
Question 4: {example_question}
[...]

Context: {example_context}
Question 1: {example_question}
Question 2: {example_question}
Question 3: {example_question}
Question 4: {example_question}
[...]

[...]

Context: {context}
Question 1:
Question 2:
Question 3:
Question 4:
[...]

```

Fig. 3. Prompt template for generating a complete set of questions using an 8-shot learning method. While a 4-shot template is presented here due to space limitations, the actual usage entails eight examples with different contexts and questions.

4.1. Method 1: generating a complete set of questions

Generating a complete set of questions from a given context is crucial for educators, as it enables comprehensive exploration and assessment of students' understanding (Al Faraby et al., 2023). This method ensures that educators cover a wide range of learning objectives and cognitive levels, facilitating a more thorough examination of the topic. Furthermore, generating a complete set of questions allows educators to cater to diverse student learning styles and preferences, promoting inclusivity, and enhancing overall learning outcomes. Overall, generating a complete set of questions from a given context empowers educators to create meaningful learning experiences that support students' holistic development and mastery of the subject matter.

To commence the evaluation of Method 1, we adopt a zero-shot learning approach, as illustrated in Fig. 2. Zero-shot prompts are formulated to stimulate the LLM into generating a complete set of questions

without explicit training. Each LLM is individually prompted with the zero-shot prompt, and the generated questions are collected and evaluated for their comprehensiveness and variety.

In addition to the zero-shot approach, we also explored 2-shot and 4-shot learning settings. In these intermediate settings, two and four examples were included in the prompt, respectively, to guide the LLM's generation of a complete set of questions. However, upon evaluation, the 2-shot and 4-shot settings did not demonstrate significant improvements in performance over the zero-shot setting. Therefore, the results of the 2-shot and 4-shot experiments have not been included in this article.

Subsequently, as depicted in Fig. 3, we proceed to the 8-shot learning method. Eight distinct examples within the prompt are provided to guide the LLM in generating a broader range of questions. The questions generated from each LLM are then assessed against predefined benchmarks to evaluate their completeness and diversity.

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Generate question at each level of Bloom's revised taxonomy.

Context: {context}
Remembering:
Understanding:
Applying:
Analyzing:
Evaluating:
Creating:

```

Fig. 4. Zero-shot prompt template for generating questions aligned with Bloom's revised taxonomy.

To ensure a fair and meaningful comparison, we provided the prompt to school teachers² (i.e., human baseline), and the entire task was divided among them, similar to that used for the LLMs, as shown in the 0-shot (Fig. 2) and 8-shot (Fig. 3) settings.

4.2. Method 2: generating questions aligned with bloom's revised taxonomy

Bloom's revised taxonomy serves as a valuable tool for educators to formulate educational questions that target specific learning objectives (Sideeg, 2016; Gul et al., 2020; Al Faraby et al., 2023). In Method 2, we concentrate on generating questions that correspond to the levels outlined in Bloom's revised taxonomy, which organizes cognitive skills into a hierarchy that includes *remembering*, *understanding*, *applying*, *analyzing*, *evaluating*, and *creating*. We begin with a zero-shot learning method, where prompts are designed to elicit questions that fit within each cognitive category of the taxonomy. Each LLM is given the zero-shot prompt (illustrated in Fig. 4), and the resulting questions are assessed for their alignment with Bloom's revised taxonomy. Instead of generating one question at a time for each taxonomic level, all six questions pertinent to a given context are produced simultaneously, as presented in Fig. 4.

Following the zero-shot method, we also explored intermediate settings, including both 2-shot and 4-shot learning. In these settings, two and four examples were provided in the prompt, respectively, to guide the LLMs in generating questions aligned with Bloom's revised taxonomy. Despite these efforts, we observed that neither the 2-shot nor the 4-shot configurations yielded significant performance improvements over the zero-shot approach. The generated questions in these settings did not show a substantial increase in alignment with the cognitive levels of Bloom's revised taxonomy, either qualitatively or quantitatively. Consequently, the results of these experiments were not included in this article, as they did not contribute to a meaningful enhancement in performance.

We then proceeded to the 8-shot learning method, where eight distinct examples were provided within the prompt, each specifically crafted to target a different cognitive level of Bloom's revised taxonomy (as depicted in Fig. 5). The generated questions are assessed to determine the degree of alignment with Bloom's revised taxonomy using both quantitative measures and qualitative assessment.

4.3. Evaluation

In this section, we detail the evaluation criteria employed in our study to assess the applicability of our two varieties of automatically generated questions within a real-world academic context. To develop these criteria, we conducted consultations with high school teachers and reviewed existing literature (Nenkova & Passonneau, 2004; Mulla & Gharpure, 2023; Steuer et al., 2021; Maity et al., 2024a, 2024b). This collaborative approach ensured that our evaluation metrics were grounded in practical teaching experiences and pedagogical standards.

- (i) **Coverage:** This indicator is quantified on a scale of 1 to 5, measuring the extent to which the generated questions encompass the essential elements of the provided context. Coverage is assessed by comparing the generated questions to the pertinent information within the context, specifically by identifying key details that should be addressed. A higher score reflects a broader range of relevant information covered by the questions, aligning with the educational objectives outlined in the context (Nenkova & Passonneau, 2004).
- (ii) **Grammaticality:** This criterion is evaluated numerically on a scale from 1 to 5, focusing on the adherence of the generated questions to standard grammatical rules and conventions. Evaluators examine the linguistic structure, syntax, and clarity of the questions to determine their grammatical correctness. A score of 5 signifies that the question is free of grammatical errors, while lower scores reflect varying degrees of grammatical issues (Maity et al., 2024a, 2024b).
- (iii) **Usefulness:** The usefulness of a question is rated on a scale from 1 to 5, indicating its potential value for inclusion in a teacher's assessment (Elkins et al., 2023). This metric evaluates whether a question can effectively contribute to student learning and assessment outcomes. It is important to note that a question does not need to be fully answerable based on context to be deemed useful. Similar types of metrics have previously been applied by Steuer et al. (2021) and Mulla and Gharpure (2023) on various scales.
- (iv) **Relevance:** This indicator assesses the semantic alignment between each generated question and the corresponding input context, also rated on a scale from 1 to 5. A question is deemed relevant if it directly relates to the content and themes presented in the context. Evaluators consider whether the question maintains focus on key concepts and avoids introducing unrelated ideas. This evaluation method is consistent with approaches used in previous studies, where both binary and scaled assessments have been implemented to measure relevance (Steuer et al., 2021; Mulla & Gharpure, 2023; Wang et al., 2024).
- (v) **Answerability:** It is rated on a scale from 1 to 5, used to gauge if a question can be addressed based on the provided context. Finding an exact portion in the context is not mandatory; instead, if a student could reasonably respond to the question using the logic conveyed in the context, it is considered 'answerable' (Steuer et al., 2021; Wang et al., 2024).

The scoring guidelines for each of these metrics can be found in the Appendix A.

Following Mulla and Gharpure (2023), we also employ a binary metric known as redundancy to evaluate the quality of the generated questions, in which we ask the evaluator to determine whether a generated question from a single context is redundant or not. In educational settings or assessments, redundant questions can take up valuable time without providing additional learning opportunities or insights. A set of questions that includes redundancies can dilute the overall quality of the question set, making it less engaging or useful. So, it is not a good idea to generate redundant questions. So in this study, we try to measure the percentage of redundant questions generated by the different

² We hired five certified school teachers through UpWork.

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Generate question at each level of Bloom's revised taxonomy. Please take care of all the
examples given below while generating the questions.

Context: {example_context}
Remembering: {example_question}
Understanding: {example_question}
Applying: {example_question}
Analyzing: {example_question}
Evaluating: {example_question}
Creating: {example_question}

Context: {example_context}
Remembering: {example_question}
Understanding: {example_question}
Applying: {example_question}
Analyzing: {example_question}
Evaluating: {example_question}
Creating: {example_question}

Context: {example_context}
Remembering: {example_question}
Understanding: {example_question}
Applying: {example_question}
Analyzing: {example_question}
Evaluating: {example_question}
Creating: {example_question}

Context: {example_context}
Remembering: {example_question}
Understanding: {example_question}
Applying: {example_question}
Analyzing: {example_question}
Evaluating: {example_question}
Creating: {example_question}

[...]

Context: {context}
Remembering:
Understanding:
Applying:
Analyzing:
Evaluating:
Creating:

```

Fig. 5. Prompt template for generating questions aligned with Bloom's revised taxonomy using an 8-shot learning method. While a 4-shot template is presented here due to space limitations, the actual usage entails eight examples with different contexts and questions.

LLMs to check which LLM generates the highest and lowest percentages of redundant questions.

This metric assesses whether two questions elicit the same answer without requiring the student to engage in different perspectives or thought processes. To determine whether questions are redundant, we asked the human evaluator to check the following: First, both questions must lead to the same answer, indicating they elicit identical responses from students. Second, there should be a lack of variation in perspective, meaning the questions should not prompt students to approach the answer from different angles or viewpoints. Finally, the cognitive processes involved in answering both questions should be similar or nearly the same, ensuring that the thought process required to arrive at the answer remains consistent. These criteria enable a comprehensive evaluation of redundancy in the generated questions. The percentage of redundant questions is calculated using the formula:

$$P_{\text{redundant}} = \left(\frac{N_{\text{redundant}}}{N_{\text{total}}} \right) \times 100 \quad (1)$$

The symbols used in the formula are defined as follows: $P_{\text{redundant}}$ represents the percentage of redundant questions generated from a single context, $N_{\text{redundant}}$ denotes the number of redundant questions as determined by the human evaluator from a single context, and N_{total} indicates the total number of questions generated from a single context.

To evaluate redundancy, we first compute the percentage of redundant questions for each context using the formula presented in Equation (1). These individual percentages are then averaged across all contexts to obtain an overall measure.

4.4. Recruitment of teacher evaluators

To enhance the credibility of our evaluation process, we provided transparency in the recruitment of our teacher evaluators. We hired a total of sixteen school teachers through the [UpWork](#) platform, selecting candidates based on their experience in high school education, familiarity with the subject matter, and a minimum of ten years of teaching experience. These educators were tasked with scoring the questions generated by both human experts (using Method 1 outlined in Section 4.1) and LLMs (using Method 1 outlined in Section 4.1) on the aforementioned scale of 1 to 5 across the five evaluation metrics, according to the scoring guideline mentioned in Appendix A. To ensure a comprehensive evaluation, we calculated the average scores assigned to all questions for each specific human evaluation metric, comparing both the LLM-generated and human-generated questions.

Furthermore, for the redundancy metric, we asked the same set of evaluators to assess whether an LLM-generated question (using Method 1 outlined in Section 4.1) from a single context was redundant. They were instructed to mark a question as '0' if deemed redundant and as

Table 1

Results of human evaluation of questions generated by 0-shot prompting of LLMs (Method 1).

Model	Coverage	Grammaticality	Usefulness	Relevance	Answerability
Human Baseline	4.89	4.95	4.46	4.83	4.68
GPT-4 Turbo	4.88	4.82	3.89	4.01	3.42
GPT-3.5 Turbo	4.68	4.78	3.67	3.89	3.23
Llama-2-70B	4.59	4.66	3.65	3.73	3.12
Llama-3.1-405B	4.74	4.80	3.80	3.95	3.30
Gemini Pro	4.67	4.75	3.68	3.90	3.25

'1' if it was not. For each context, we calculate the percentage of redundant questions as described in Equation (1) and subsequently average these values across all contexts. This systematic approach allows us to quantify redundancy effectively and assess the diversity of the questions generated.

In addition to the sixteen evaluators, we recruited another twelve school teachers, also through UpWork. These teachers, each with at least ten years of teaching experience, were responsible for categorizing the questions generated by the LLMs (using Method 2 outlined in Section 4.2) into six distinct categories: *understanding*, *remembering*, *applying*, *analyzing*, *evaluating*, and *creating*. This classification task was systematically divided among the twelve teachers, ensuring that each category received focused attention.

This structured approach to evaluation not only provides a robust assessment of the generated questions but also reinforces the credibility of our findings by incorporating insights from practicing educators.

5. Results and discussion

As illustrated in Table 1, the human baseline consistently outperforms all LLMs across the five evaluation metrics, *coverage*, *grammaticality*, *usefulness*, *relevance*, and *answerability*, demonstrating the superiority of human-generated questions in educational contexts. Among the LLMs, GPT-4 Turbo emerges as the best performer, particularly excelling in *coverage* and *grammaticality*, though it still falls short in terms of *usefulness* and *answerability* compared to the human baseline. Llama-3.1-405B closely follows GPT-4 Turbo, performing well in *grammaticality* and *coverage*. However, like GPT-4 Turbo, it struggles with creating *answerable* questions from contexts. GPT-3.5 Turbo and Gemini Pro are closely matched in performance, both showing moderate success in *coverage* and *grammaticality*. However, their generated questions fall further behind in terms of *relevance* and *answerability*, highlighting limitations in how they comprehend and process the input context. Llama-2-70B, while capable, ranks the lowest among the evaluated LLMs, consistently underperforming in most categories. This model's difficulties, particularly in generating questions that are *useful* and *answerable*, underscore the challenges of relying on less advanced models for complex educational tasks.

Following Chiang and Lee (2023), we similarly computed the Kendall's correlation coefficient (τ) to examine the alignment between the evaluations given by top-performing LLM (i.e., GPT-4 Turbo) and the human evaluators. Kendall's correlation coefficient is an important metric for measuring the alignment between the evaluations given by GPT-4 Turbo and human evaluators. Overall, using Kendall's correlation coefficient helps to validate the performance of LLMs like GPT-4 Turbo in generating content that meets human evaluators' standards. By comparing the scores given by GPT-4 Turbo and human evaluators, Kendall's coefficient provides a quantitative measure of how closely the model's output aligns with human judgments.

According to Table 2,³ the Kendall's correlation coefficient (τ) for GPT-4 Turbo indicates a strong correlation in terms of *coverage* and

Table 2The Kendall's correlation coefficient (τ) between the best-performing LLM (i.e., GPT-4 Turbo) and human evaluators in the 0-shot setting (Method 1).

Criterion	GPT-4 Turbo
Coverage	0.32
Grammaticality	0.34
Usefulness	0.26
Relevance	0.25
Answerability	0.21

grammaticality for Method 1 under the 0-shot setting. However, there is only a moderate correlation between human experts and GPT-4 Turbo in terms of *usefulness*, *relevance*, and *answerability*.

As seen in Table 3, the overall performance of all LLMs improved under the 8-shot prompting setting when compared to the 0-shot results. However, the human baseline continues to achieve the highest scores across all evaluation metrics except *coverage* for Method 1 under the 8-shot setting. Among the LLMs, GPT-4 Turbo remains the best performer, showing notable improvements across most metrics, particularly in *usefulness* and *relevance*. The increase in examples (i.e., 8-shot) allowed GPT-4 Turbo to generate more contextually aligned and practical questions, narrowing the gap between it and human evaluators, though it still lags behind in *answerability*. Llama-3.1-405B also saw significant gains in *usefulness* and *relevance*, positioning itself close to GPT-4 Turbo in these areas. This indicates that providing more examples during prompting helps Llama-3.1-405B generate questions that are more aligned with educational standards, though it still struggles to match the human baseline in *answerability*. GPT-3.5 Turbo and Gemini Pro show moderate improvements, particularly in *relevance* and *usefulness*, indicating their ability to benefit from the additional context provided through 8-shot prompting. However, both models continue to fall short in terms of generating fully *answerable* questions compared to their more advanced counterparts. Llama-2-70B, while improving from its 0-shot performance, still ranks at the bottom. Despite some gains in *grammaticality* and *relevance*, its questions remain less *useful* and *answerable* than those produced by other models. This suggests that even with more examples, Llama-2-70B struggles to effectively capture the complexity of educational question generation.

In summary, the 8-shot setting enhances the performance of all LLMs, with GPT-4 Turbo and Llama-3.1-405B benefiting the most. However, none of the LLMs fully bridge the gap to human performance, particularly in terms of question *usefulness* and *answerability*.

However, as illustrated in Table 4, in the 8-shot setting (Method 1), the Kendall's (τ) for the best-performing LLM, GPT-4 Turbo, has improved across all human evaluation metrics, indicating a stronger correlation between human judgments and those of GPT-4 Turbo compared

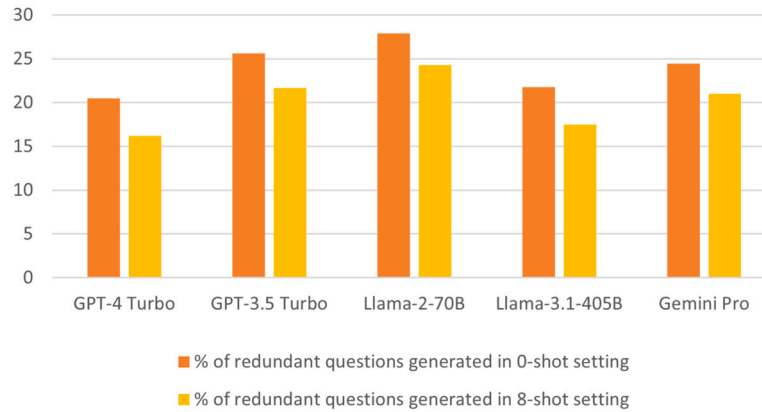
³ When interpreting Kendall's τ , $|\tau| \in [0, 0.1)$ is considered as very weak correlation, $|\tau| \in [0.1, 0.2)$ is considered as weak correlation, $|\tau| \in [0.2, 0.3)$ is

considered as moderate correlation and $|\tau| \in [0.3, 1.0]$ is considered as strong correlation (Botsch, 2011).

Table 3

Results of human evaluation of questions generated by 8-shot prompting of LLMs (Method 1).

Model	Coverage	Grammaticality	Usefulness	Relevance	Answerability
Human Baseline	4.89	4.95	4.46	4.83	4.68
GPT-4 Turbo	4.90	4.88	4.10	4.57	3.88
GPT-3.5 Turbo	4.86	4.82	3.88	4.45	3.84
Llama-2-70B	4.62	4.78	3.75	4.32	3.62
Llama-3.1-405B	4.88	4.86	4.03	4.53	3.85
Gemini Pro	4.85	4.83	3.86	4.44	3.86

**Fig. 6.** Comparison of the percentage of redundant questions generated by five LLMs in 0-shot and 8-shot settings for Method 1.**Table 4**The Kendall's correlation coefficient (τ) between the best-performing LLM (i.e., GPT-4 Turbo) and human evaluators in the 8-shot setting (Method 1).

Criterion	GPT-4 Turbo
Coverage	0.41
Grammaticality	0.38
Usefulness	0.33
Relevance	0.34
Answerability	0.28

to the 0-shot setting. In the 8-shot setting, GPT-4 Turbo shows a strong correlation with human experts across all human evaluation metrics except for *answerability*, where the correlation is moderate.

One critical aspect of the evaluation was the *usefulness* of the generated questions, which human evaluators rated lower than expected, particularly in the 0-shot setting (Table 1). This low rating could be attributed to the model's struggle to capture nuanced educational objectives, especially without the examples provided in few-shot settings. In 0-shot settings, the models generate questions based on pre-trained knowledge alone, often resulting in questions that are *grammatically* correct but lack depth or pedagogical focus. To improve *usefulness*, one strategy could be refining the prompts used in question generation. By providing more examples, as we have done in the 8-shot setting (Table 3), models can generate questions that are better aligned with educational objectives.

Fig. 6 shows the comparison of percentage of *redundant* questions generated by five LLMs in 0-shot and 8-shot settings for Method 1. We observe that when we use the 8-shot setting, the percentage of *redundant* questions generated decreases compared to that of the 0-shot setting for Method 1.

According to the prompt provided in Method 2, the LLMs should generate questions distributed equally across the six categories of Bloom's

Taxonomy, with each category comprising 16.67% of the generated questions. Table 5 illustrates that in the 0-shot setting (Method 2), the distribution of questions across Bloom's revised taxonomy categories varies significantly from the expected 16.67%. However, in Table 6, after presenting 8-shot examples, the percentage of questions generated for each category aligns more closely with the target of 16.67%, demonstrating the effectiveness of providing 8-shot examples. Additionally, among the LLMs, GPT-4 Turbo and Llama-3.1-405B produce questions that most closely match the ideal distribution of 16.67% for each question category according to Bloom's revised taxonomy.

We also discuss a few generated questions from two subjects, economics, and history, using Method 1 and Method 2 in 0-shot and 8-shot settings.

Method 1 (0-shot setting, Table 7) reveals that certain questions were repetitive. Specifically, Questions 2 and 9, as well as Questions 4 and 7, repeated the same concepts, leading to *redundancy*. Furthermore, Question 10 contained a *grammatical error*, where the indefinite article 'a' was missing from the phrase "fully accurate comparison". The correct phrasing should be: "Why might PPP not always provide a fully accurate comparison of two countries' economies?" However, the 8-shot setting (Table 8) for Method 1 significantly reduced these issues. None of the questions exhibited *redundancy*, and *grammatical errors* were not present, highlighting the benefit of including more contextual examples for the model to learn from.

In Method 2 (0-shot setting, Table 9), we observed the misclassification of questions according to Bloom's revised taxonomy. For instance, a question categorized as 'evaluating' should have been classified as 'analyzing' and vice versa. These discrepancies were corrected in the 8-shot setting (Table 10), where all generated questions were accurately classified into their respective categories based on Bloom's taxonomy.

A similar pattern emerged in history questions. In Method 1 (0-shot setting, Table 11), *redundancy* was again an issue. Questions 2 and 3 referred to the same concept, and Question 10 was *irrelevant* to the context. These problems were resolved in the 8-shot setting (Table 12), where all questions were contextually appropriate and none were *redundant*.

Finally, for Method 2 (0-shot setting, Table 13), a question intended to assess 'applying' was better suited to 'analyzing' according to Bloom's

Table 5

Percentage of questions generated according to different Bloom's revised taxonomy levels via 0-shot prompting of LLMs (Method 2).

Model	Understanding	Remembering	Applying	Analyzing	Evaluating	Creating
GPT-4 Turbo	25.62	18.39	15.42	12.47	14.20	13.88
GPT-3.5 Turbo	27.52	20.42	18.32	11.61	8.92	13.20
Llama-2-70B	29.17	20.49	20.52	10.66	7.94	11.20
Llama-3.1-405B	26.55	19.59	14.58	12.34	13.55	13.37
Gemini Pro	28.86	21.04	17.63	10.42	8.98	13.04

Table 6

Percentage of questions generated according to different Bloom's revised taxonomy levels via 8-shot prompting of LLMs (Method 2).

Model	Understanding	Remembering	Applying	Analyzing	Evaluating	Creating
GPT-4 Turbo	18.62	17.39	15.89	14.47	15.20	18.40
GPT-3.5 Turbo	19.62	18.63	15.93	14.61	12.92	18.25
Llama-2-70B	20.27	19.52	19.95	14.44	12.12	13.63
Llama-3.1-405B	18.69	17.55	15.80	14.40	15.25	18.29
Gemini Pro	19.60	18.45	16.07	14.45	13.34	18.05

Table 7

A Context from the EduProbe dataset (Maity et al., 2024a) related to Economics, accompanied by questions produced by GPT-4 Turbo using Method 1 in 0-shot setting.

Context (Economics): Purchasing power parity (PPP) is an economic indicator that signifies the purchasing power of the currencies of various nations of the world against each other. It helps in comparing living standards between different countries and estimating economic productivity.

- Question 1: What is Purchasing Power Parity (PPP) and how is it used in economic analysis?
 Question 2: How does PPP help in comparing the living standards between different countries?
 Question 3: What could be some limitations or challenges of using PPP as a measure of economic productivity?
 Question 4: How does PPP contribute to global economic predictions and analyses of international finance?
 Question 5: How does PPP adjust for the differences in price levels between countries?
 Question 6: In what ways do governments or international organizations use PPP data in policy-making?
 Question 7: What role does PPP play in global economic forecasts and international financial analyses?
 Question 8: How is PPP different from nominal Gross Domestic Product (GDP) per capita in assessing a country's economic performance?
 Question 9: How does PPP assist in the comparison of living standards across various nations?
 Question 10: Why might PPP not always provide fully accurate comparison of two countries' economies?

Table 8

A Context from the EduProbe dataset (Maity et al., 2024a) related to Economics, accompanied by questions produced by GPT-4 Turbo using Method 1 in 8-shot setting.

Context (Economics): Purchasing power parity (PPP) is an economic indicator that signifies the purchasing power of the currencies of various nations of the world against each other. It helps in comparing living standards between different countries and estimating economic productivity.

- Question 1: What is Purchasing Power Parity (PPP) and how is it calculated?
 Question 2: How does PPP differ from nominal exchange rates when comparing the economic strength of countries?
 Question 3: What are some examples of how PPP is used to adjust the GDP of countries for more accurate comparisons?
 Question 4: Can you explain the limitations or criticisms associated with using PPP as an economic indicator?
 Question 5: How does PPP contribute to understanding global income inequalities?
 Question 6: What role does PPP play in international trade and investment decisions?
 Question 7: How frequently is PPP data updated, and which organizations are responsible for its calculation?
 Question 8: How can fluctuations in PPP affect the perceived living standards between different countries?
 Question 9: What are the challenges in measuring PPP among countries with significant differences in economic structure or consumption patterns?
 Question 10: How has the concept of PPP evolved over time in economic theory and practice?

revised taxonomy. In the 8-shot setting (Table 14), however, the generated questions aligned correctly with Bloom's revised taxonomy, demonstrating a significant improvement in the precision of categorization.

The results provide clear guidance for educators seeking to use AI for automated question generation. For Method 1, the 8-shot setting consistently outperformed the 0-shot setting across all metrics, including *coverage*, *grammaticality*, *usefulness*, *relevance*, *answerability*, and *redundancy*. Additionally, for Method 2, the 8-shot setting outperforms the 0-shot setting in generating questions aligned with Bloom's revised taxonomy. This suggests that when using LLMs for the generation of educational questions, educators should provide more examples in the prompt (as in the 8-shot setting) while generating the questions to ensure higher-quality outcomes.

6. Conclusion

Our study explores the potential of leveraging LLMs for automated question generation in the educational domain. We focus on generating school-level questions using context extracted from NCERT textbooks across several subjects. By harnessing the capabilities of state-of-the-art LLMs, we prompt them to generate a *complete set of questions* from contexts extracted from textbooks under 0-shot and 8-shot settings. This is followed by a human evaluation of the generated questions in terms of *coverage*, *grammar*, *usefulness*, *answerability*, and *relevance*. In addition, we also prompt LLMs to generate questions based on Bloom's revised taxonomy and then analyze these questions through human experts in both 0-shot and 8-shot settings.

Table 9

A Context from the EduProbe dataset (Maity et al., 2024a) related to Economics, accompanied by questions produced by GPT-4 Turbo using Method 2 in 0-shot setting.

Context (Economics): Purchasing power parity (PPP) is an economic indicator that signifies the purchasing power of the currencies of various nations of the world against each other. It helps in comparing living standards between different countries and estimating economic productivity.

Remembering: What does PPP stand for in economic terms?
 Understanding: Explain how purchasing power parity helps in comparing living standards between different countries.
 Applying: How might an economist use PPP to analyze the economic situation of a developing country compared to a developed country?
 Analyzing: Assess the effectiveness of PPP as an economic indicator when considering its application in global financial markets.
 Evaluating: Examine the limitations of using PPP as a measure for comparing economic productivity between two vastly different economies.
 Creating: Propose a new model or index that could complement or improve upon PPP for assessing international economic comparisons.

Table 10

A Context from the EduProbe dataset (Maity et al., 2024a) related to Economics, accompanied by questions produced by GPT-4 Turbo using Method 2 in 8-shot setting.

Context (Economics): Purchasing power parity (PPP) is an economic indicator that signifies the purchasing power of the currencies of various nations of the world against each other. It helps in comparing living standards between different countries and estimating economic productivity.

Remembering: What does PPP stand for in economic terms?
 Understanding: How does purchasing power parity help in comparing living standards between countries?
 Applying: How would you use PPP to determine if it's more economical to live in Country A compared to Country B?
 Analyzing: Analyze the limitations of using PPP as a measure for comparing economic productivity across different countries.
 Evaluating: Evaluate the effectiveness of PPP as a reliable indicator for economic planning and policy-making. What are its strengths and weaknesses?
 Creating: Devise a tool or software that can dynamically calculate and compare the PPP of different countries, taking into account changes in currency value and inflation rates.

Table 11

A Context from the EduProbe dataset (Maity et al., 2024a) related to History, accompanied by questions produced by GPT-4 Turbo using Method 1 in 0-shot setting.

Context (History): During the medieval period in India, Islamic rulers held significant power, leading to the blending of Indian and Islamic cultures, which can still be observed in the architecture and artwork created at that time. The country was governed and administered by notable rulers who made exceptional contributions in the fields of art, culture, literature, and architecture during this period.

Question 1: Who were some of the prominent Islamic rulers in India during the medieval period, and what were their major contributions to art and culture?
 Question 2: How did the architectural styles in India evolve during the medieval period under Islamic influence?
 Question 3: How did the architectural designs in India change during the medieval era with the influence of Islamic culture?
 Question 4: How did Islamic rulers impact the administration and governance of India during the medieval period?
 Question 5: Can you discuss the role of literature in cultural exchange between Indian and Islamic traditions during this era?
 Question 6: What were the social and economic effects on Indian society due to the rule of Islamic leaders in the medieval period?
 Question 7: How did religious practices and philosophies integrate or conflict during the reign of Islamic rulers in medieval India?
 Question 8: What preservation efforts exist today for the architectural and artistic works from the medieval period in India?
 Question 9: How have modern scholars and historians interpreted the influence of Islamic rulers on medieval Indian society?
 Question 10: What were the significant occurrences in India during World War (WW2)?

Table 12

A Context from the EduProbe dataset (Maity et al., 2024a) related to History, accompanied by questions produced by GPT-4 Turbo using Method 1 in 8-shot setting.

Context (History): During the medieval period in India, Islamic rulers held significant power, leading to the blending of Indian and Islamic cultures, which can still be observed in the architecture and artwork created at that time. The country was governed and administered by notable rulers who made exceptional contributions in the fields of art, culture, literature, and architecture during this period.

Question 1: Who were some of the prominent Islamic rulers in India during the medieval period, and what were their contributions to art and culture?
 Question 2: In what ways did Indian and Islamic cultures blend during the medieval period in India?
 Question 3: How did Islamic rulers influence the architecture in India during the medieval period?
 Question 4: Can you describe specific examples of artwork from the medieval period in India that reflect the blend of Indian and Islamic cultures?
 Question 5: What impact did Islamic rule have on the literature of India during the medieval period?
 Question 6: How did the administrative practices of Islamic rulers in India during the medieval period affect the local populace?
 Question 7: What are some of the lasting impacts of the Islamic rule on modern Indian society and culture?
 Question 8: How did religious practices and beliefs change in India under the influence of Islamic rulers?
 Question 9: What were the economic impacts of Islamic rule in India during the medieval period?
 Question 10: How has the historical interpretation of the Islamic period in Indian history evolved over time in academic and popular discourse?

Table 13

A Context from the EduProbe dataset (Maity et al., 2024a) related to History, accompanied by questions produced by GPT-4 Turbo using Method 2 in 0-shot setting.

Context (History): During the medieval period in India, Islamic rulers held significant power, leading to the blending of Indian and Islamic cultures, which can still be observed in the architecture and artwork created at that time. The country was governed and administered by notable rulers who made exceptional contributions in the fields of art, culture, literature, and architecture during this period.

Remembering: What are some examples of architectural styles that emerged during the medieval period in India under Islamic rulers?
Understanding: How did the blend of Indian and Islamic cultures influence the artwork and architecture developed during the medieval period in India?
Applying: How would you differentiate between purely Indian and Indo-Islamic architectural features if you were to explain it to someone unfamiliar with the topic?
Analyzing: What were the political, social, and economic factors that led to the blending of Indian and Islamic cultures during the medieval period in India?
Evaluating: Assess the contributions of Islamic rulers to Indian art and culture. Were these influences predominantly positive, negative, or a mix of both?
Creating: Design an educational exhibit for a museum that showcases the fusion of Indian and Islamic cultures in medieval India, including sections on architecture, literature, and art.

Table 14

A Context from the EduProbe dataset (Maity et al., 2024a) related to History, accompanied by questions produced by GPT-4 Turbo using Method 2 in 8-shot setting.

Context (History): During the medieval period in India, Islamic rulers held significant power, leading to the blending of Indian and Islamic cultures, which can still be observed in the architecture and artwork created at that time. The country was governed and administered by notable rulers who made exceptional contributions in the fields of art, culture, literature, and architecture during this period.

Remembering: What is an example of architecture from the medieval period in India that shows the blending of Indian and Islamic cultures?
Understanding: How did the governance by Islamic rulers during the medieval period influence cultural developments in India?
Applying: How would you use the historical impact of Islamic rulers' contributions to art and culture to design a curriculum about medieval Indian history?
Analyzing: Analyze the effects of Islamic rulers' policies on the literature produced during the medieval period in India.
Evaluating: Evaluate the overall impact of Islamic rule on Indian architecture. Do you think the contributions were beneficial in the long-term cultural fabric of India?
Creating: Develop a presentation that highlights the integration of Indian and Islamic cultures during the medieval period, featuring examples of architecture, art, and literature that emerged from this fusion.

We observed that LLMs are able to generate a *complete set of questions* from a particular context related to the school curriculum and that the quality of the generated questions improves in the 8-shot setting over the 0-shot setting. Moreover, presenting 8-shot examples reduces the *redundancy* of LLM-generated questions from the same context compared to the 0-shot setting. Furthermore, LLMs generate questions according to Bloom's revised taxonomy more effectively in the 8-shot setting than in the 0-shot setting. Overall, we observe that LLMs have reached close to human-level performance in generating school-level questions by applying the *few-shot learning*. The findings suggest that automated question generation from school-level textbooks using LLMs holds promise to assist educators in creating diverse and high-quality questions. For practical implications, educators can utilize the insights from this study to enhance their question-generation practices, particularly by incorporating example prompts to improve the quality of LLM-generated questions. In future research, we plan to explore 16-shot and 32-shot settings to assess whether increasing the number of examples further enhances the quality of the LLM-generated questions.

CRedit authorship contribution statement

Subhankar Maity: Writing – review & editing, Writing – original draft, Software, Resources, Methodology. **Aniket Deroy:** Writing – review & editing, Software, Resources, Methodology. **Sudeshna Sarkar:** Writing – review & editing, Supervision, Methodology.

Statements on open data and ethics

The data used in this study were sourced from publicly available datasets and literature. No private or confidential data were accessed or utilized. The study was approved by the Institutional Ethics Committee (IEC) in accordance with the relevant ethical guidelines and regulations established by the committee. No personally identifiable or sensitive information was collected, ensuring complete anonymity for

the participants. Comprehensive details about the study were shared with participants before the study commenced.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to proofread, polish, and enhance the clarity and style of their initial drafts. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Scoring guidelines for evaluation metrics

In this appendix, we provide the scoring guidelines used to evaluate the quality of the generated questions based on the defined human evaluation metrics (See Section 4.3). Each metric is rated on a scale from 1 to 5, with specific criteria for each score.

(i) Coverage

- 1: Very few or no relevant elements covered from the context.
- 2: Some elements are covered from the context, but key details are missing.
- 3: Several important elements are covered from the context, but some details are overlooked.
- 4: Broad coverage of essential elements from the context.
- 5: Comprehensive coverage of all essential elements from the context.

- (ii) *Grammaticality*
 - 1: Numerous errors; difficult to understand.
 - 2: Several errors; meaning is somewhat unclear.
 - 3: Some errors; generally understandable.
 - 4: Minor errors; mostly clear and correct.
 - 5: No errors; completely clear and correct.
- (iii) *Usefulness*
 - 1: Not useful; no contribution to learning.
 - 2: Limited usefulness; minimal impact on assessment.
 - 3: Moderately useful; contributes to learning, though somewhat predictable.
 - 4: Useful; enhances both assessment and student understanding.
 - 5: Highly useful; significantly boosts learning and assessment outcomes.
- (iv) *Relevance*
 - 1: Not relevant; no connection to the context.
 - 2: Slightly relevant; weak connection to the context.
 - 3: Moderately relevant; some focus on the context.
 - 4: Relevant; closely aligns with the context.
 - 5: Highly relevant; directly addresses the context without unrelated ideas.
- (v) *Answerability*
 - 1: Not answerable; lacks any basis in the context.
 - 2: Difficult to answer; weak connection to the context.
 - 3: Moderately answerable; some logic can be applied from the context.
 - 4: Mostly answerable; clear basis in the context.
 - 5: Fully answerable; directly supported by the context.

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